

D281

# Linux Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

On a web server in a Virginia data center, a 2 KB backup script must let its owner read, write, and run it, let the group read and run it, and give others no access. Which command sets this?

- A. `chmod 750 backup.sh`
- B. `chmod 570 backup.sh`
- C. `chmod 740 backup.sh`
- D. `chmod 755 backup.sh`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

`ls -l` shows `-rw-r--r--` for a file. What is its octal mode?

- A. 744
- B. 622
- C. 466
- D. 644

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

On a laptop running Ubuntu 24.04 with 40 GB free, `pwd` prints `/home/analyst/projects`. The user runs `cd ../docs`. What does `pwd` print now?

- A. `/home/analyst/docs`
- B. `/home/analyst/projects/docs`
- C. `/home/docs`
- D. `/docs`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A cron job runs at midnight: `grep error app.log > errors.txt`. The file `app.log` is 900 MB and did not change between runs. After the job runs on two nights in a row, what does `errors.txt` contain?

- A. Two copies of the matching lines
- B. The matches plus `grep`'s error messages
- C. One copy of the matching lines
- D. Nothing, since `>` always empties the file

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

Which command adds today's matching lines to the end of `errors.txt`, keeps its old contents, and hides `grep`'s error messages?

- A. `grep error app.log | errors.txt`
- B. `grep error app.log > errors.txt 2>/dev/null`
- C. `grep error app.log >> errors.txt`
- D. `grep error app.log >> errors.txt 2>/dev/null`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A company with 200 engineers modifies a GPL-licensed media library and ships it inside a product it sells for \$49. What does the license require?

- A. Only credit the original authors.
- B. Release the source code of the modified library under the GPL to recipients.
- C. Stop selling; the GPL forbids charging money.
- D. Nothing, because it charges a fee.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

A developer wants others to use her library in closed-source products as long as they keep her copyright notice. Which license fits?

- A. Any open-source license, since they all allow this
- B. A non-commercial Creative Commons license
- C. MIT
- D. GPL

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

An admin must install nginx on a Rocky Linux server with 16 GB of RAM that hosts a bakery's website. Which command fits?

- A. `sudo dnf install nginx`
- B. `sudo apt install nginx`
- C. Compile from source, since servers can never use package managers
- D. `sudo dpkg -i nginx.deb`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

An auditor reviewing a server with 120 user accounts asks where the hashed passwords are stored. Which file should the admin point to?

- A. `/etc/group`
- B. `/etc/passwd`
- C. `/etc/shadow`
- D. `/var/log/auth.log`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

A mail server that processes 10,000 messages a day is rejecting mail. Where should the admin look first for the system's log files?

- A. `/tmp`
- B. `/var/log`
- C. `/etc`
- D. `/bin`

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

A file lives at `/etc/ssh/sshd_config`. What kind of file is it most likely to be?

- A. A user's personal SSH settings
- B. System-wide configuration for the SSH server
- C. An SSH log file
- D. The SSH program itself

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

An analyst needs one gzip-compressed archive of the reports directory, which holds 1,500 PDF files, to email to an auditor. Which command creates it?

- A. tar -xzf reports.tar.gz reports/
- B. tar -cf reports.tar.gz reports/
- C. gzip reports/
- D. tar -czf reports.tar.gz reports/

| Target word | The one fact that decides | Answer | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/> guess |
|-------------|---------------------------|--------|----------------------------------------------------------------------------------------------|
| _____       | _____                     | ___    |                                                                                              |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer A Concept 1

On a web server ~~in a Virginia data center~~, a ~~2-KB~~ backup script must let its owner read, write, and run it, let the group read and run it, and give others no access. Which command sets this?

**Target word:** octal mode **Decides:** rwx = 7, r-x = 5, --- = 0.

|   |                    |                                  |
|---|--------------------|----------------------------------|
| A | <b>Correct</b>     | Owner 7, group 5, others 0.      |
| B | Inverted operation | Swaps owner and group digits.    |
| C | Adjacent category  | Group gets read but not execute. |
| D | Dropped element    | Gives others read and execute.   |

## 2. Answer D Concept 1 · transfer

Is -l shows -rw-r--r-- for a file. What is its octal mode?

**Target word:** rwx string to octal **Decides:** rw- = 6, r-- = 4, r-- = 4.

|   |                    |                             |
|---|--------------------|-----------------------------|
| A | Confusion set      | Adds execute for the owner. |
| B | Adjacent category  | Reads r as 2 instead of 4.  |
| C | Inverted operation | Reverses the class order.   |
| D | <b>Correct</b>     | 6, 4, 4.                    |

## 3. Answer A Concept 2

On a laptop ~~running Ubuntu 24.04 with 40 GB free~~, pwd prints /home/analyst/projects. The user runs cd ../docs. What does pwd print now?

**Target word:** resulting directory **Decides:** .. goes to /home/analyst, then into docs.

|   |                   |                              |
|---|-------------------|------------------------------|
| A | <b>Correct</b>    | Up one level, then docs.     |
| B | Dropped element   | Ignores the ..               |
| C | Confusion set     | Goes up two levels.          |
| D | Adjacent category | Treats the path as absolute. |

## 4. Answer C Concept 3

A cron job ~~runs at midnight~~: grep error app.log > errors.txt. ~~The file app.log is 900 MB~~ and did not change between runs. After the job runs on two nights in a row, what does errors.txt contain?

**Target word:** file contents after two runs **Decides:** > overwrites each time.

|   |                   |                                                     |
|---|-------------------|-----------------------------------------------------|
| A | Confusion set     | That is >> behavior.                                |
| B | Adjacent category | > captures stdout only; errors need 2>.             |
| C | <b>Correct</b>    | The second run replaces the first.                  |
| D | Absolute language | It empties the file and then writes the new output. |

## 5. Answer D Concept 3 · transfer

Which command adds today's matching lines to the end of errors.txt, keeps its old contents, and hides grep's error messages?

**Target word:** append and discard stderr **Decides:** >> appends; 2>/dev/null discards errors.

|   |                    |                                     |
|---|--------------------|-------------------------------------|
| A | Confusion set      | A pipe feeds a command, not a file. |
| B | Inverted operation | Overwrites old contents.            |
| C | Dropped element    | Error messages still appear.        |
| D | <b>Correct</b>     | Appends stdout and discards stderr. |

## 6. Answer B Concept 4

A company ~~with 200 engineers~~ modifies a GPL-licensed media library and ships it inside a product ~~it sells for \$49~~. What does the license require?

**Target word:** GPL obligation **Decides:** Distributed modified GPL code.

|   |                   |                                               |
|---|-------------------|-----------------------------------------------|
| A | Adjacent category | That is a permissive license like MIT.        |
| B | <b>Correct</b>    | Copyleft passes the same freedoms downstream. |
| C | Confusion set     | Selling GPL software is allowed.              |
| D | Camouflage        | Price does not change GPL duties.             |

## 7. Answer C Concept 4 · transfer

A developer wants others to use her library in closed-source products as long as they keep her copyright notice. Which license fits?

**Target word:** pick license from requirement **Decides:** Permissive with attribution.

|   |                    |                                                  |
|---|--------------------|--------------------------------------------------|
| A | Absolute language  | Copyleft licenses do not.                        |
| B | Confusion set      | Blocks commercial use and is not meant for code. |
| C | <b>Correct</b>     | Permissive; requires only the notice.            |
| D | Inverted operation | Copyleft would force derived products open.      |

## 8. Answer A Concept 5

An admin must install nginx on a Rocky Linux server ~~with 16 GB of RAM that hosts a bakery's website~~. Which command fits?

**Target word:** package command for the distro **Decides:** Rocky is Red Hat-family.

|   |                   |                                                     |
|---|-------------------|-----------------------------------------------------|
| A | <b>Correct</b>    | dnf manages RPM packages on Red Hat-family systems. |
| B | Confusion set     | apt is for Debian and Ubuntu.                       |
| C | Absolute language | Package managers are the normal method.             |
| D | Adjacent category | dpkg installs .deb files, not RPMs.                 |

## 9. Answer C Concept 6

An auditor reviewing a server ~~with 120 user accounts~~ asks where the hashed passwords are stored. Which file should the admin point to?

**Target word:** location of hashes **Decides:** Hashes are kept in a root-only file.

|   |                   |                                                 |
|---|-------------------|-------------------------------------------------|
| A | Adjacent category | Lists groups and members.                       |
| B | Confusion set     | Holds account info; the password field shows x. |
| C | <b>Correct</b>    | Holds hashes and is readable only by root.      |
| D | Dropped element   | Logs login attempts, not stored hashes.         |

## 10. Answer B Concept 7

A mail server ~~that processes 10,000 messages a day~~ is rejecting mail. Where should the admin look first for the system's log files?

**Target word:** log location **Decides:** Filesystem hierarchy standard.

|   |                    |                                              |
|---|--------------------|----------------------------------------------|
| A | Adjacent category  | Temporary files that may be cleared.         |
| B | <b>Correct</b>     | Variable data such as logs lives under /var. |
| C | Confusion set      | Configuration, not logs.                     |
| D | Inverted operation | Programs, not their output.                  |

## 11. Answer B Concept 7 · transfer

A file lives at /etc/ssh/sshd\_config. What kind of file is it most likely to be?

**Target word:** infer type from location **Decides:** /etc holds system configuration.

|   |                    |                                         |
|---|--------------------|-----------------------------------------|
| A | Adjacent category  | Those live in ~/.ssh.                   |
| B | <b>Correct</b>     | /etc is for configuration.              |
| C | Confusion set      | Logs go in /var/log.                    |
| D | Inverted operation | Programs live in /usr/bin or /usr/sbin. |

## 12. Answer D Concept 8

An analyst needs one gzip-compressed archive of the reports directory, ~~which holds 1,500 PDF files,~~ to email to an auditor. Which command creates it?

**Target word:** create compressed tar **Decides:** Create + gzip + file name.

|   |                    |                                                                 |
|---|--------------------|-----------------------------------------------------------------|
| A | Inverted operation | x extracts.                                                     |
| B | Dropped element    | No -z, so it is not compressed despite the name.                |
| C | Adjacent category  | gzip compresses single files, not a directory into one archive. |
| D | <b>Correct</b>     | c creates, z compresses, f names the file.                      |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 5             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 0             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 10            |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 3             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 10            |             |
| Camouflage         | Answers a stem fact that was never the question.             | 1             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                   | Rule                                                                                                                                                   | Trap                                                                                   | Tell                                              |
|---|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------|
| 1 | <b>Octal permissions</b>                  | Each of owner, group, others is a digit: r = 4, w = 2, x = 1, summed.                                                                                  | Swapping the order of the three classes or leaving out a bit.                          | rwX strings or a chmod number.                    |
| 2 | <b>Absolute and relative paths</b>        | Paths starting with / are absolute; others start from the current directory, and .. means the parent.                                                  | Ignoring .. or treating a relative name as absolute.                                   | pwd output followed by cd with .. or a bare name. |
| 3 | <b>Redirection and pipes</b>              | > overwrites a file, >> appends, 2> redirects errors, and   sends one command's output into another command.                                           | Using > when you meant to keep old contents.                                           | Output going to a file across repeated runs.      |
| 4 | <b>Open-source licenses</b>               | Copyleft licenses (GPL) require distributed modifications to be released under the same license; permissive licenses (MIT, BSD) need only attribution. | Thinking charging money exempts you from the GPL, or that all licenses are permissive. | Modified code being shipped to customers.         |
| 5 | <b>Distributions and package managers</b> | Debian and Ubuntu use apt and .deb packages; Red Hat, Fedora, and Rocky use dnf and .rpm packages.                                                     | Running apt on a Red Hat-family system.                                                | A named distribution and an install task.         |
| 6 | <b>User account files</b>                 | /etc/passwd holds account info and is world-readable; hashed passwords live in root-only /etc/shadow.                                                  | Thinking passwords are stored in /etc/passwd today.                                    | "Where are the password hashes?"                  |
| 7 | <b>Filesystem hierarchy</b>               | /etc = system configuration, /var/log = logs, /home = user files, /tmp = temporary files, /bin and /usr/bin = programs.                                | Looking in /etc for logs.                                                              | A kind of file and "where would it be?"           |
| 8 | <b>Archiving with tar</b>                 | tar -c creates, -x extracts, -z adds gzip compression, and -f names the archive file.                                                                  | Using -x to create, or forgetting -z for a compressed archive.                         | Bundle and compress a whole directory.            |

**Course scope (public description):** Linux Foundations prepares learners for the LPI Linux Essentials certification, and is an introduction to Linux as an operating system as well as an introduction to open-source concepts and the basics of the Linux command line. Learners will gain skills in identifying the fundamentals of open-source software and to develop resources for data access and security.